Problem Chosen	2025 MCM/ICM Summary Sheet	Team Control Number
D		2504020

Links among Individuals: An Optimized Higher-order Transportation Network Model

The development of transportation has facilitated the movement of people and brought significant economic benefits. For Baltimore, however, due to aging infrastructure and limited transportation options, people's lives have not been able to keep up with the pace of progress, exacerbated by the collapse of the bridge. The purpose of this report is to develop an optimized higher-order transportation network model to assess the local impacts of transportation planning.

To build the model, we first perform data cleaning. To ensure the uniqueness of each road node, we merge several geographic information fields to generate a composite ID, and filter out some nodes with missing information. Next, we construct a directed graph, filter the effective directed edge set according to the above method, and then use traffic flow data to assign weights to each directed edge. Then, we integrate the data, construct higher-order nodes based on the idea of **higher-order network model (HON)**, and combine the set of directed edges to generate the local traffic network topology. Then, based on the **cascade failure model**, parameters such as node capacity are introduced to estimate the chain reaction formed by the failure of nodes, which may lead to the failure of other nodes and edges in complex networks. Finally, the node importance evaluation system is constructed, and the indicators are selected by considering the network structure and the role of information transfer between nodes, and the entropy weight **TOPSIS method** is adopted to quantitatively estimate the importance of nodes based on their relative proximity to the ideal solution.

For task I, we first rank the known nodes based on the established model and node importance evaluation metrics. Next, the bridge collapse is modeled abstractly as the disappearance of nodes of a certain size (accompanied by the elimination of associated directed edges) in a higher-order network model. Then, by selecting suitable nodes close to the bridge as benchmarks, attacking the selected set of nodes, adjusting the affected failed nodes using the cascade failure model, and visualizing the results in the **three-tiered transportation network** corresponding to different interest groups (cf. Figure 14-19), we can then analyze the impact of the bridge collapse on the local transportation ecosystems and different populations.

For task II and task III, we studied "The West Baltimore United (WBU) Project" and constructed an undirected graph model using 2,722 bus stations as nodes and 394 bus routes as edges. We defined the weights of nodes and edges, incorporating inflow and outflow data as evaluation metrics. Preliminary analysis revealed route overlap in the urban area and insufficient coverage in the suburbs. **Community detection** using **modularity optimization algorithms** showed poor connectivity between urban and suburban communities. As a result, we proposed an optimization plan: integrating redundant urban routes to improve efficiency, and designing new cross-regional rapid transit routes to enhance suburban connectivity. The implementation is divided into three phases, with an expected 30% improvement in community connectivity, a 25% increase in operational efficiency, and a 35% reduction in suburban commute time after optimization.

Keywords: HON; Cascade Failure Model; TOPSIS; Three-tiered Transportation Network

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1 Introduction

1.1 Problem Background

Baltimore, located in the state of Maryland in the United States, is a vibrant and historic metropolis. With a rich maritime heritage, it has a bustling port that has been crucial for trade and commerce for centuries.

Baltimore has a well-developed transportation system. It has a public transport network including buses, light rail and metro. There is also an extensive network of roads and highways, such as Interstate 95, which runs through the city, making it easily accessible to those driving to other major cities on the East Coast. However, the problems of aging traffic facilities and limited traffic lines have gradually been exposed with the change of history.

However, there is no end to transport reform. The collapse of the Francis Scott Key Bridge laid bare the city's deep-rooted weaknesses in the planning of its transport network. How to optimize the traffic system so that everyone who lives in it can get satisfaction and happiness has become an unavoidable problem. We needed to find a solution that would improve the connection between people, communities, and cities in Baltimore.

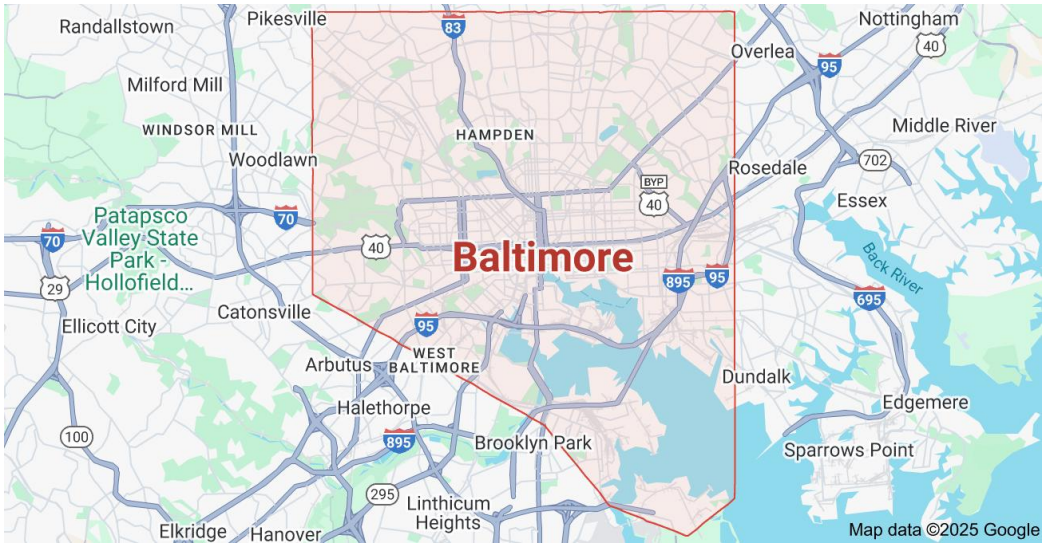


Figure 1: Baltimore City Map

1.2 Our work

The topic requires us to establish a local transportation network model for Baltimore based on the given dataset, and discuss the impacts of different events on local residents and other relevant stakeholders according to the established model. Our work mainly includes:

- ✧ Based on the given local traffic dataset, a local transportation network model for Baltimore was established.
- ✧ Improve the evaluation system for the network model, introduce different parameters to

✧ conduct multi - dimensional quantitative evaluations of the model's simulations under different hypothetical scenarios.

In summary, the whole modeling process can be shown as follows:

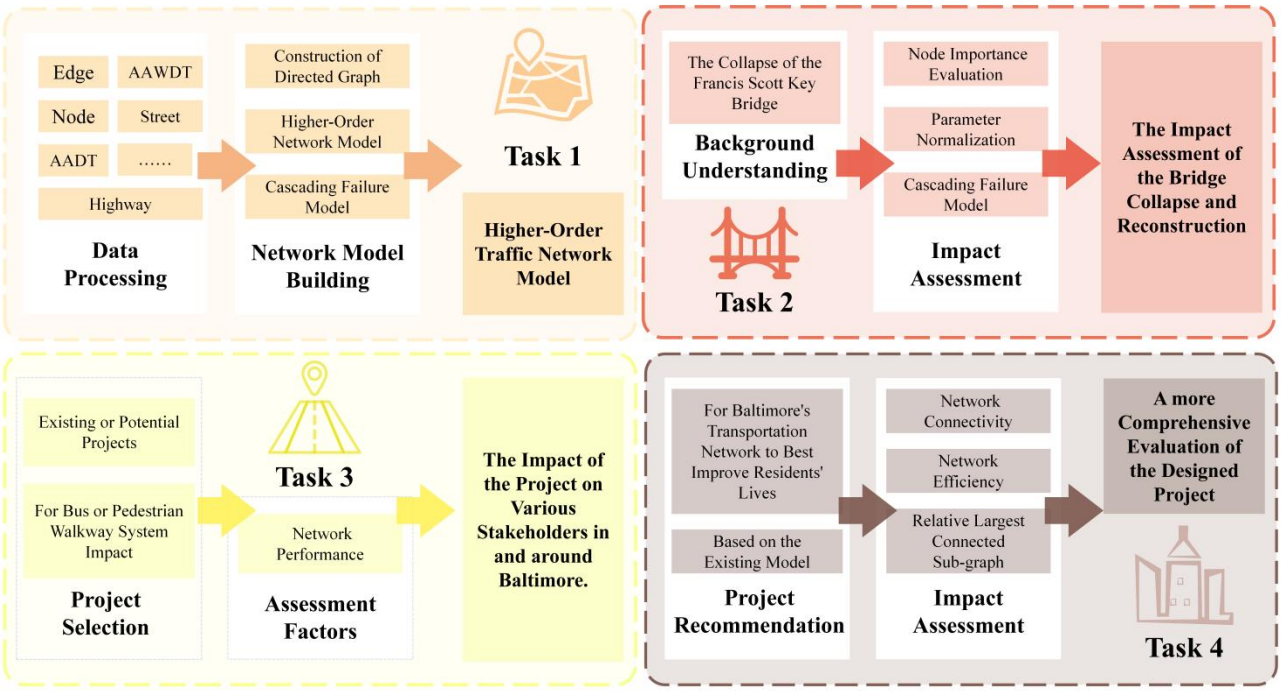


Figure 2: Our Work

2 Assumptions and Justifications

To simplify the problem, we make the following basic assumptions, each of which is properly justified.

➢ **Assumption 1: The size of each node and each path can be ignored.**

Justification: The core of transportation network models tends to be the study of topological properties such as the connectivity between nodes and the paths along which traffic flows run, while the actual sizes of the nodes and roads have little impact on the understanding of such topologies and route planning. Neglecting their sizes allows the model to present the essential features of the network more clearly, which facilitates studies such as topology analysis and route planning.

➢ **Assumption 2: The traffic on each path is evenly distributed between each edge of that path.**

Justification: Since roadway topologies can be relatively complex and traffic flow data is collected at the path level rather than at the specific edge level, uniform distribution simplifies the flow delivery process.

➢ **Assumption 3: The weights of the edges are directly associated to each edge through**

the traffic flow data.

Justification: To facilitate subsequent traffic flow analysis and optimization, traffic flow data (e.g., AADT) are bound directly to the edges of the traffic network as edge weights, providing the basis for subsequent quantitative traffic flow assessment and route optimization problems.

- **Assumption 4: Assume the research data is accurate.**

Justification: We assume that the data given in the question related to the local transportation system in Baltimore is accurate and true.

3 Notations

The key mathematical notations used in this paper are listed in Table 1.

Table 1: Notations used in this paper

Symbol	Description
w_{ji}	The weight of the edge directed from node j to node i
$S_{out}(j)$	The out-degree of node j
$S_{in}(j)$	The in-degree of node j
σ_{st}	The number of shortest paths from node s to node t
$\sigma_{st}(i)$	The number of paths passing through node i among the number of shortest paths from node s to node t
k_i	The total number of nodes directly connected to node i in the network
N	The number of nodes in the network
d_{ij}	The shortest path from node i to node j
V	Network node set

4 Model Preparation

4.1 The Data

Since the amount of data is large and not intuitive, we directly visualize some of the data for display.

4.1.1 Data Collection

The data we use mainly come from the datasets provided in the compressed package and the relevant open - data provided by the Baltimore County Government. The data sources are summarized in Table 2.

Table 2: Data source collation

Database Names	Database Websites Data	Type
Local Roads	https://arcg.is/05489r2	Geography
Alleys and Driveways	https://arcg.is/1PXKbr0	Geography

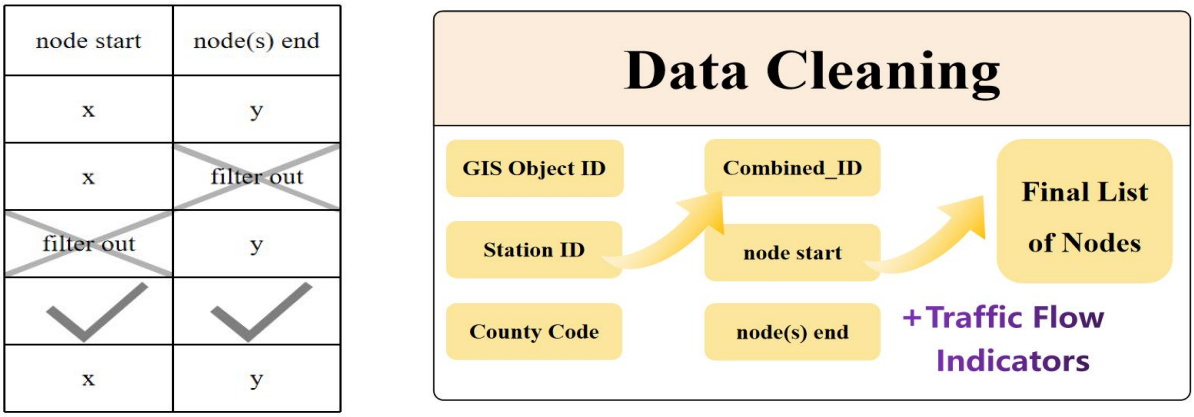
4.1.2 Data Cleaning

First, to ensure the uniqueness of each road node, we merge multiple geographic information fields to generate a composite ID, which is then merged into the data file, providing a traceable node identifier for the subsequent construction of directed edges.

Key attributes directly related to traffic flow were then filtered out, including traffic flow metrics (e.g., AADT) and topological information. Records that did not have valid topological relationships (the information about the node was null) were deleted, ensuring that only rows of data that could be analyzed topologically were retained, allowing us to focus on data that was directly relevant to traffic flow analysis.

The list is then traversed to exclude invalid nodes, which in turn ensures that we are able to comprehensively capture all possible nodes in the data and ultimately construct the complete set of nodes.

The specific workflow can be seen in the following figure:



(a) Node Data Filtering (b) Data Cleaning Process

Figure 3:Data Cleaning

5 Model: Higher-Order Traffic Network Model

Urban road transportation is complex, not only a variety of types, but also for each

individual road to measure its indicators are quite a lot. Therefore, such a complex network system can be solved by constructing an advanced network model. Formally, directed graphs well simulate the directionality and complexity of the traffic network, and maximize the connotation of the data without introducing exogenous factors. Therefore, the abstract simulation based on higher-order network models can well approximate the local traffic network in Baltimore.

5.1 Construction of Directed Graphs

In the previous steps of data cleaning, we have completed the selection of nodes. But in the process of actually constructing a directed graph, in addition to completing the selection of nodes, we also need to complete the selection of directed edges.

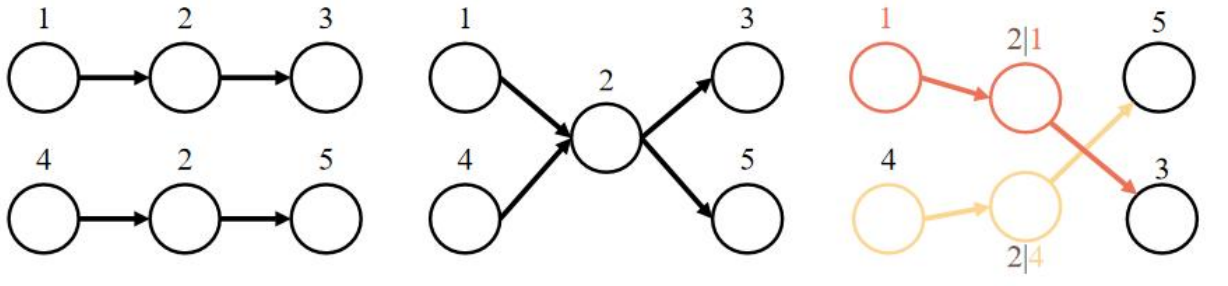
We utilize the node start and node(s) end columns of the filtered dataset to define the directed edges in this step. Specifically, the starting point in node start (hereafter abbreviated as s) of each record is combined with each end point in node(s) end (hereafter abbreviated as e) to form a directed edge $(s \to e)$.

For the further refinement of the traffic flow analysis, we also use the traffic flow data (e.g., AADT) as the weights of the edges. Through this process, we generate the full set of directed edges, which provides structured network data for subsequent traffic flow analysis and optimization.

5.2 Higher-Order Network Model

When mathematically modeling the traffic network, the pairwise relational model and the higher-order network model are considered first. A comparison of the results of the pairwise relational model and the higher-order network model is shown in Figure 4. It can be seen from Figure 4:

Based on pairwise relational assumption modeling, it shows four direct trajectory information $1 \to 2 \to 3$, $1 \to 2 \to 5$, $4 \to 2 \to 3$, $4 \to 2 \to 5$, but in reality, the two direct routes $1 \to 2 \to 5$ and $4 \to 2 \to 3$ do not exist in reality. By using the higher-order network model, node 2 is abstracted into two higher-order nodes, $2|1$ and $2|4$, according to the difference of the antecedent nodes, and $2|1$ is connected to 1 and 3, and $2|4$ is connected to 4 and 5, which accurately expresses the information of the real traffic routes.



(a)Original route alignment (B)Pairwise relational model (C)Higher-order network model

Figure 4: Comparison of Results from Pairwise Relational Models and Higher-order Network Models[1]

Considering that traffic route data is typical sequence data with time-dependent characteristics, we choose to construct a higher-order network model based on the dataset.

5.3 Cascading Failure Model

In a complicated network, the failure of a node may lead to the failure of other nodes and edges, thus forming a chain reaction. When constructing the cascade failure model of the directed weighted subway network, it is assumed that the nodes and edges are not resilient during the test period after being attacked.

5.3.1 Node Initial Load and Capacity

According to the existing studies, considering the complexity of the structure of urban subway network, the concept of "node load" is introduced, and the node degree value is used to define the initial load of nodes.[2] The key parameters are as follows:

$$L_i = k_i^{\theta} \tag{1}$$

Where: L_i is the initial load; k_i is the total degree value of the node; θ is the adjustable parameter of the initial load.

Next, we set all nodes in the network to have the ability to bear the load, considering that the node capacity is directly proportional to the initial load of the node, and the node capacity is defined as follows:

$$C_i = T \times L_i \tag{2}$$

Where: C_i is the initial capacity of the node; T is the capacity regulation parameter.

5.3.2 Load Redistribution Model

Within a transportation network, a node failure may occur at a station due to internal or external failures. Once this occurs, its effects may spread to neighboring nodes, which in turn may trigger cascading failures. When a node fails due to a contingency, the load it originally carried will be distributed to the neighboring and unfailed nodes. If a neighboring node exceeds its capacity due to new load, a new failed node is created and the next round of load distribution is opened. When a node suffers an attack, its load is redistributed according to the established load redistribution policy. The load redistribution process is repeated until the entire network is partially or completely down.

During load redistribution, the load distribution ratio is calculated based on the weights of directed edges and the importance of nodes, taking into account the directionality of edges and the importance of nodes in the network. Assuming that the failed node in the network is i and the neighboring station is j, then the load allocated from node i to node j is calculated as follows:

$$\Delta L_j = \left(\alpha \frac{w_{ij}}{\sum_{j\in\varphi} w_{ij}} + \beta \frac{S_j}{\sum_{j\in\varphi} S_j} \right) \quad (1)$$

Where: ΔL_j denotes the load increment of node j; w_{ij} is the weight of the directed edge of node i pointing to node j; S_j is the importance of node j; φ is the set of the neighboring nodes of node i; α, β are the weights assigned based on the weights of the directed edges and the importance of the nodes, respectively, with $\alpha + \beta = 1$.

According to equations (1) ~ (2), i.e., the initial load of the node and the node capacity to determine the node state. If $L_i > C_i$, the node fails and load redistribution is performed; if $L_i \le C_i$, the node is normal and can continue to receive load.

5.3.3 Simulation Flow of Cascade Failure Algorithm

The steps of cascade failure modeling algorithm are as follows.

1) Step 1: Determine the urban transportation network, obtain the weights of each edge in the network through statistics, and form a directed weighted network.

2) Step 2: Calculate the network node degree and other characteristic indexes according to Matlab. The initial load and capacity of each node of the network are obtained according to equations (1) ~ (2).

3) Step 3: Fail a node in the network according to the attack strategy, remove the node and its neighboring edges from the network, and form a new network.

4) Step 4: Distribute the failed node load. Distribute the failed node load to the neighboring nodes according to equation (3).

5) Step 5: Judge the state of the node, for the node whose load is greater than the capacity, repeat Step 4 until there is no new failed node, otherwise the failure process ends. The flow is shown in Figure 5.

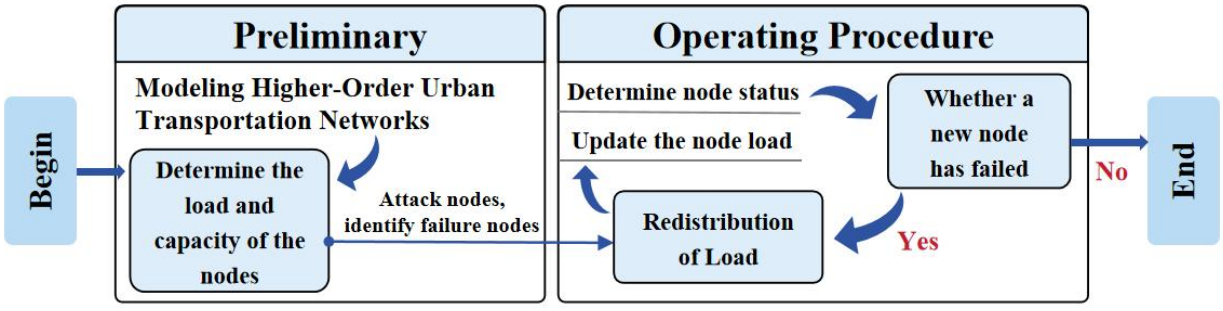


Figure 5: Transportation Network Cascade Failure Simulation Flow

The Baltimore transit network, including 730 effective nodes, is constructed based on the abstraction of the higher-order network model, and the topology of the higher-order transit network is visualized using Gephi, as shown in Figure 6.

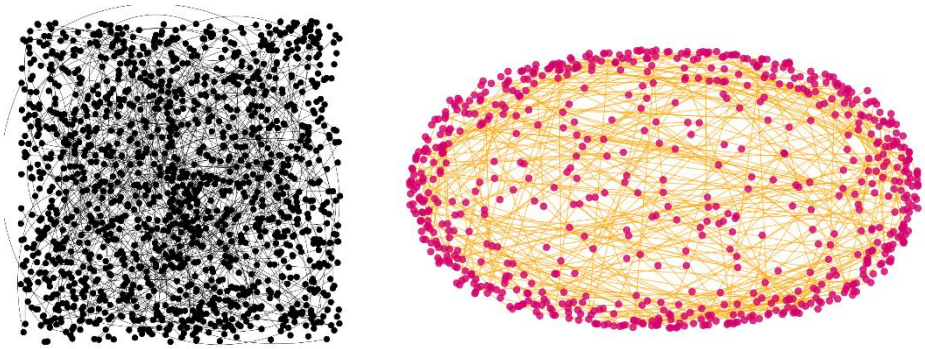


Figure 6: Baltimore Higher Order Transportation Network Topology Map

5.4 Nodal Importance Evaluation

5.4.1 Indicators for Evaluating the Importance of Nodes

In the transportation network, the failure of any critical node may trigger a chain reaction and lead to a large-scale network failure. Therefore, accurate identification of key nodes is crucial to the analysis of metro network destructiveness. The selected indicators should not only consider the network structure, but also need to take into account the role of information transfer between nodes.

In complex networks, nodes do not exist in isolation, but are related to each other. Based on this, the node attraction rate AR and node transfer rate TR can be used to measure the degree of node's own contribution to other nodes, as well as other node eigenvalues to assess the degree of node transfer information.[2]

1) Nodal attraction rate AR, which is calculated as shown in equation (4).

$$AR = \sum_{j=1}^{n} \frac{w_{ji}}{S_{out}(j)} \tag{1}$$

2) Node transfer rate TR, which is calculated as shown in equation (5).

$$TR = \sum_{s=1}^{n} \frac{w_{is}}{S_{in}(s)} \tag{2}$$

3) Betweenness centrality B_i, which is calculated as shown in equation (6).

$$B_i = \sum_{s \neq i \neq t} \frac{\sigma_{st}(i)}{\sigma_{st}} \tag{3}$$

4) Degree ki, which is calculated as shown in equation (7).

$$k_i = \sum_{j=1}^{n} a_{ij} \tag{4}$$

Where: k_i denotes the total number of nodes in the network that are directly connected to node i; a_{ij} denotes that if site i is connected to site j, then $a_{ij} = 1$, otherwise, $a_{ij} = 0$.

5) The characteristic path length L, which is calculated as shown in equation (8).

$$L = \frac{1}{N(N-1)} \sum_{i,j \in V, i \neq j} d_{ij} \tag{5}$$

Where: N denotes the number of nodes in the network; V denotes the set of nodes in the network.

5.4.2 Methods for identifying key nodes

The entropy weight TOPSIS method fully considers the correlation between the indicators and the allocation of weights, which can efficiently deal with the problem of ambiguity and uncertainty, and at the same time, it considers the distance between the indicator values and the ideal solution and the worst solution, and the results are intuitive and easy to interpret. Therefore, we choose to use the entropy weight TOPSIS method to determine the weights of the indicators, evaluate the importance of the nodes, and rank the importance of the nodes accordingly. [3]

The calculation steps are as follows:

1) Computing key node evaluation metrics for networks.

2) Standardize the indicators to obtain a standardized evaluation matrix, as shown in the following equations:

$$P = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mn} \end{bmatrix} \tag{1}$$

$$p_{ij} = x_{ij} \Big/ \sqrt{\sum_{i=1}^{m} x_{ij}^2} \tag{2}$$

Where: s_{ij} denotes the standardized data of the corresponding indicator; x_{ij} denotes the original data of the jth indicator of the i-th node.

3) Under the jth indicator, the value of the ith program of the weight of the indicator P_{ij}, which is calculated as shown in equation (11).

$$P_{ij} = p_{ij} \Big/ \sum_{i=1}^{m} p_{ij} \left(j = 1, 2, \cdots, n \right) \tag{3}$$

4) Calculate the entropy value Ej of the jth indicator, as shown in equation (12).

$$E_j = -\frac{1}{\ln n} \sum_{i=1}^{m} P_{ij} \ln P_{ij} \left(j = 1, 2, \cdots, n \right) \tag{4}$$

5) Calculate the coefficient of variation G_j for the jth indicator, which is calculated as shown in equation (13).

$$G_j = 1 - E_j \tag{5}$$

6) Calculate the weight value of the jth indicator W_j, which is calculated as shown in equation (14).

$$W_j = G_j \Big/ \sum_{j=1}^{n} G_j \tag{6}$$

7) Construct the weighting matrix and multiply each column of the normalized matrix Z with the corresponding weights.

8) Determine the positive ideal solution Z_+ and the negative ideal solution Z_-.

9) Calculate the distance from the object to be evaluated to the positive ideal solution D_i^+ and the distance from the negative ideal solution D_i^-.

10) Use the negative ideal solution distance to calculate the relative proximity of the evaluation object and the optimal solution S_i. S_i value is between 0 and 1, the higher the value indicates that the node has a higher degree of importance. The calculation is shown in equation (15).

$$S_i = \frac{D_i^-}{D_i^+ + D_i^-} \tag{7}$$

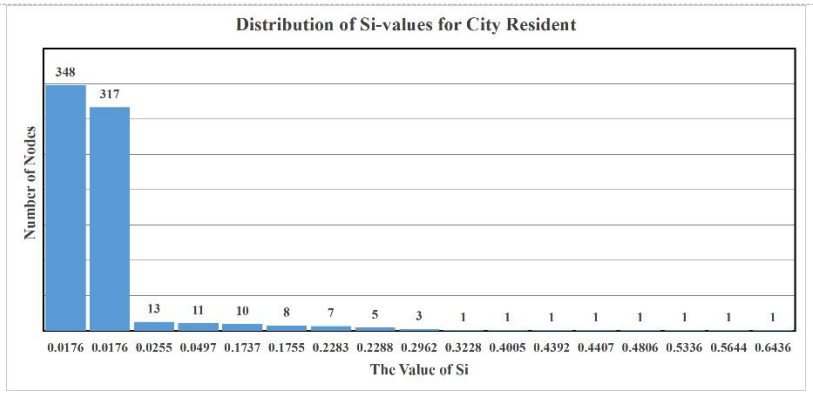
11) Sort the nodes in descending order of importance based on the value of Si.

6 Task I: the Collapse of Francis Scott Key Bridge

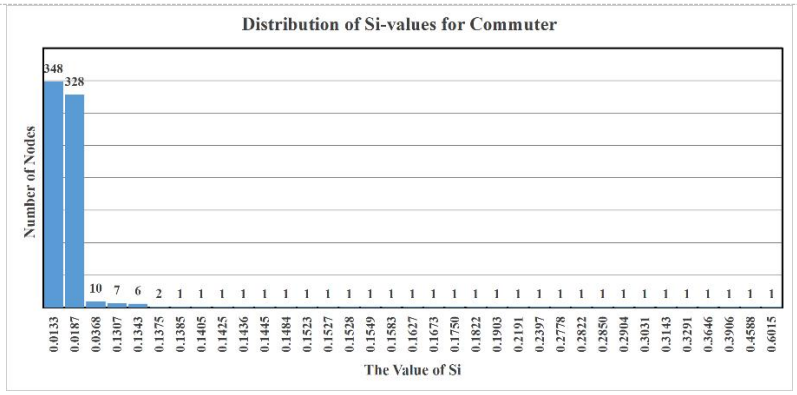
On the basis of the above established model and the node importance evaluation index, we can rank the known nodes. The collapse of the bridge can be modeled as the disappearance of nodes of a certain size (accompanied by the elimination of the directed edges to which they are connected) in the established higher-order network model. For directed graphs, the significance of node disappearance is thin. However, the value of a single node or a cluster of nodes is not the same for different interest groups, so it is not possible to measure the significance of the bridge for them by a set of criteria. To account for such differences, we can change the weights of the indicators in the judgment matrix to better simulate their actual interests, and finally measure the value of the bridge to the local people of Baltimore.

6.1 Node Importance Ranking

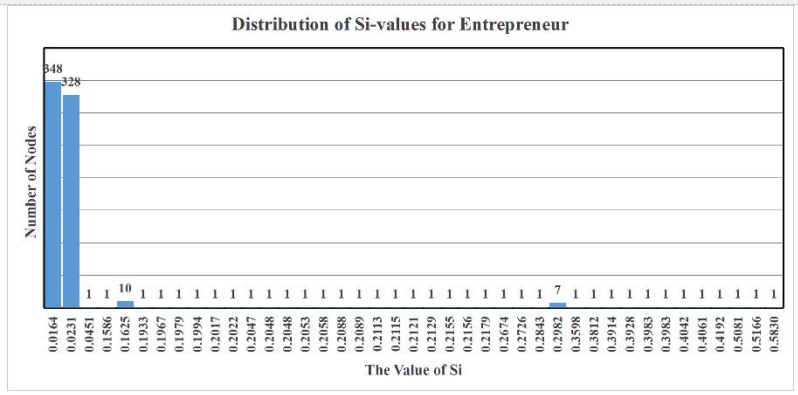
Based on the improved entropy weight TOPSIS method, the S_i values of all nodes in the Baltimore transportation network for different interest groups are calculated and the data distribution results are shown in Figure 6:[4]



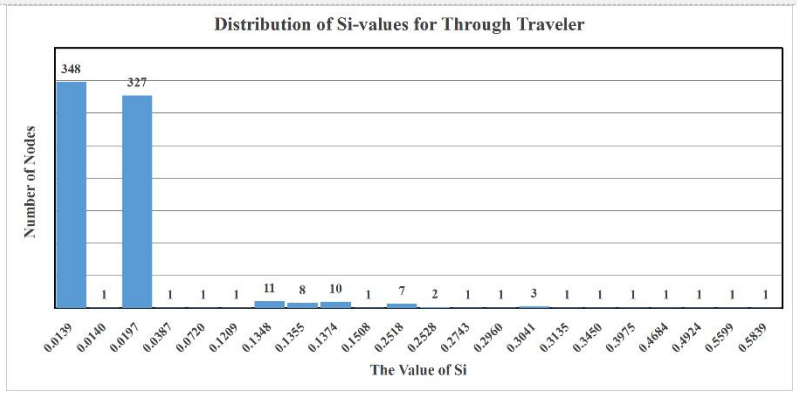
(a) Distribution of Si-values for City Resident



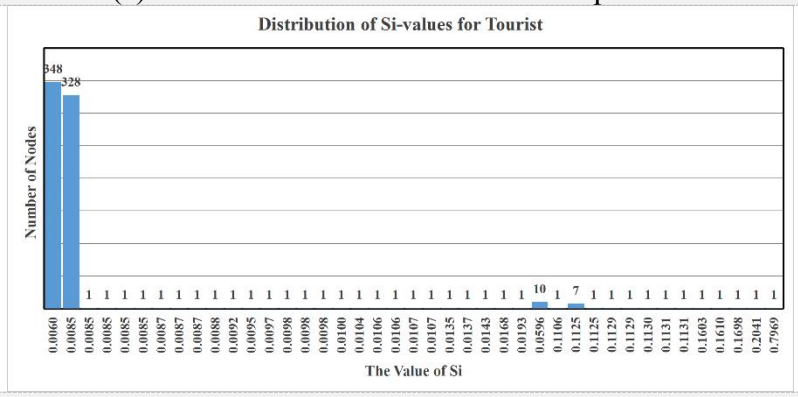
(b) Distribution of Si-values for Commuter



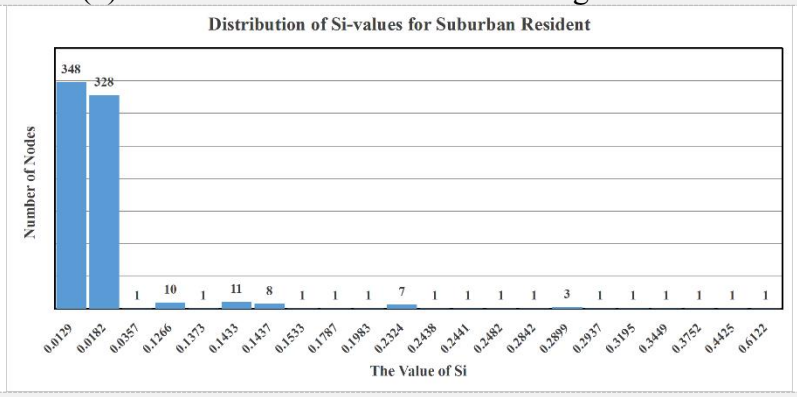
(c) Distribution of Si-values for Entrepreneur



(d) Distribution of Si-values for Through Traveler



(e) Distribution of Si-values for Tourist



(e) Distribution of Si-values for Suburban Resident

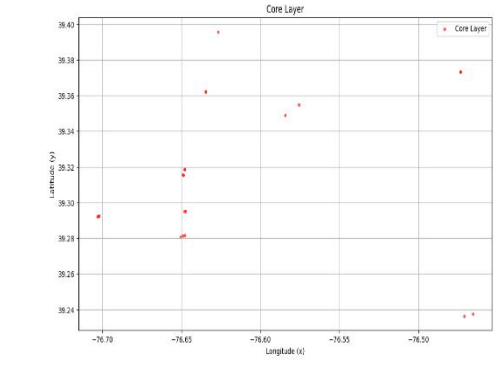
Figure 7: Distribution of Si values for different population nodes in transportation networks

6.2 Nodal Multi-Layer Structure Division

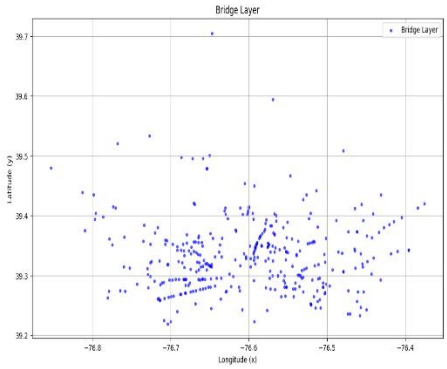
Consider the data distribution of Si values to stratify the higher order transportation network.

Firstly, the data of S_i values are analyzed, if at $S_i < x$, the value distribution of nodes in the network is always continuous, and at x the distribution of Si values is broken and there is a gap, so the nodes with $S_i \geq x+1$ are defined as the core layer. Secondly, analyzing the data of low values, according to the given nodes and value distribution, it can be concluded that the difference of the cumulative share of nodes if peaked at $S_i \leq y$, so the nodes with $S_i \leq y-1$ are defined to constitute the peripheral layer. Finally, the nodes with residual values between $y \leq S_i \leq x-1$ are defined as the bridge layer in the higher order transit network.

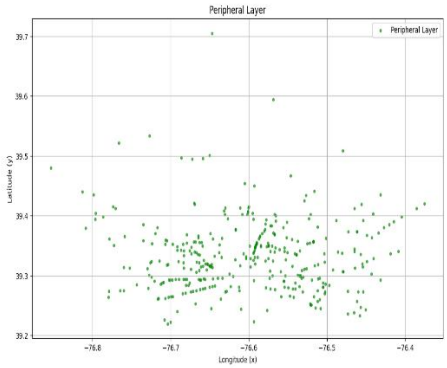
Based on the above layering method, the high-order transit network is divided into three different layers, and the topology of the layering is shown in Figure 8-13.



(a)Core Layer

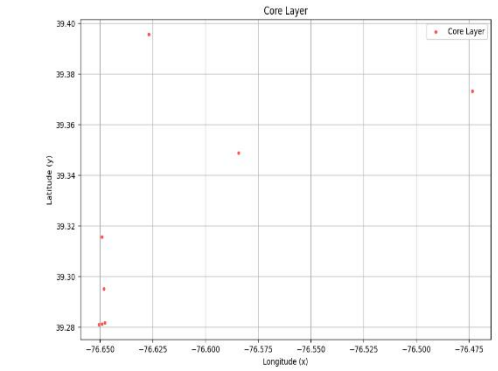


(b)Bridge Layer

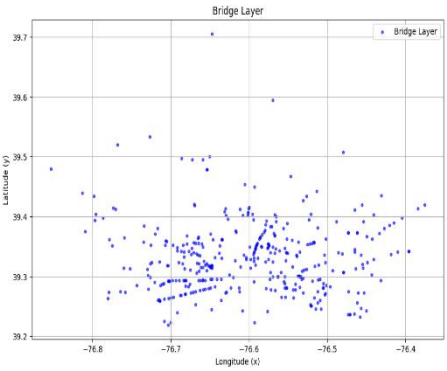


(c)peripheral layer

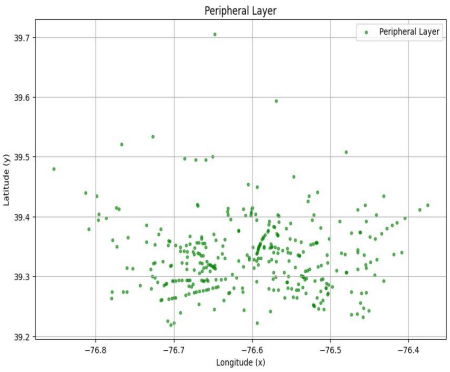
Figure 8: Three-layer topology of a high-order transit network (for city resident)



(a)Core Layer

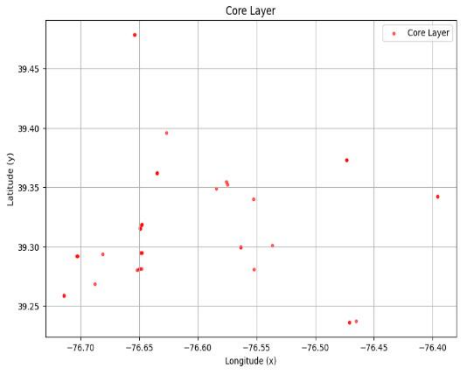


(b)Bridge Layer

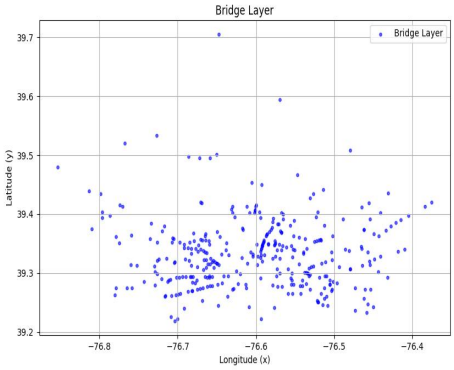


(c)peripheral layer

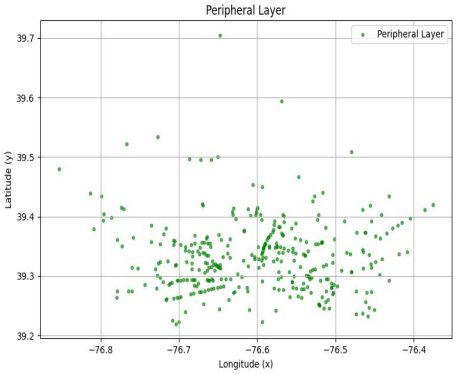
Figure 9: Three-layer topology of a high-order transit network (for commuter)



(a)Core Layer

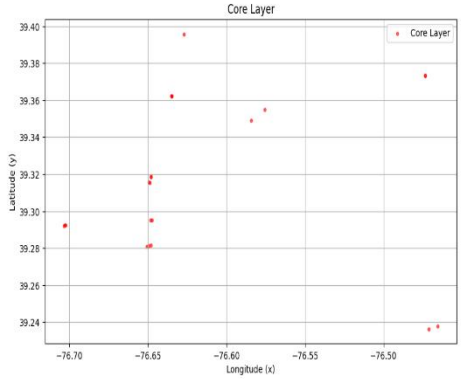


(b)Bridge Layer

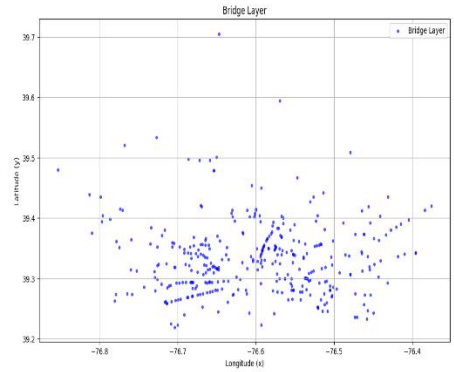


(c)peripheral layer

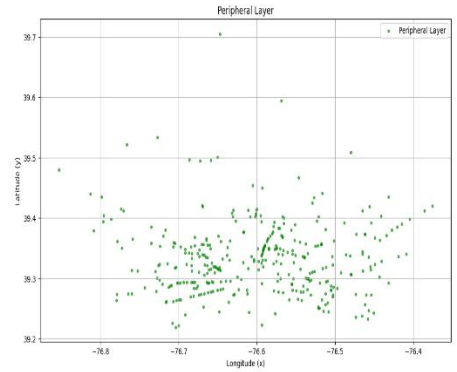
Figure 10: Three-layer topology of a high-order transit network (for entrepreneur)



(a)Core Layer

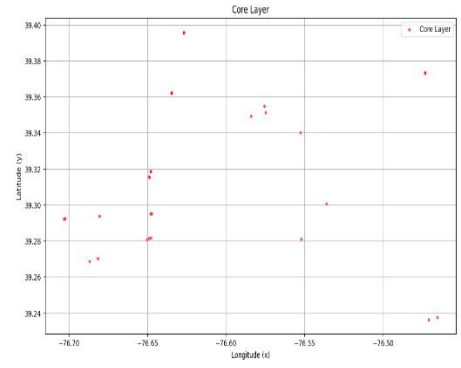


(b)Bridge Layer

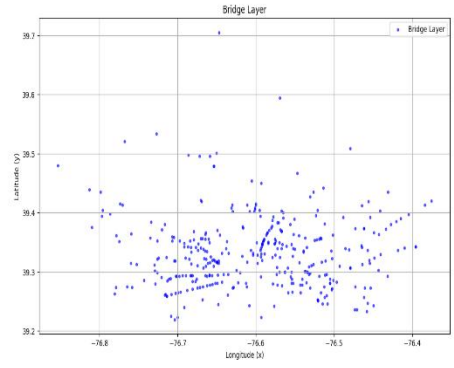


(c)peripheral layer

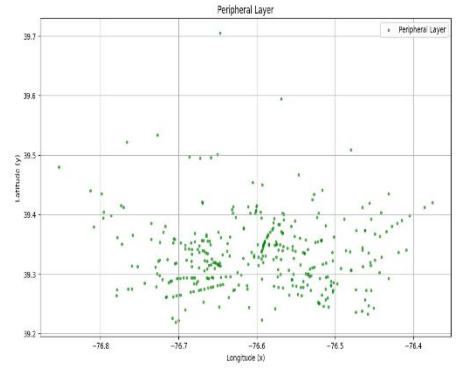
Figure 11: Three-layer topology of a high-order transit network (for through traveler)



(a)Core Layer

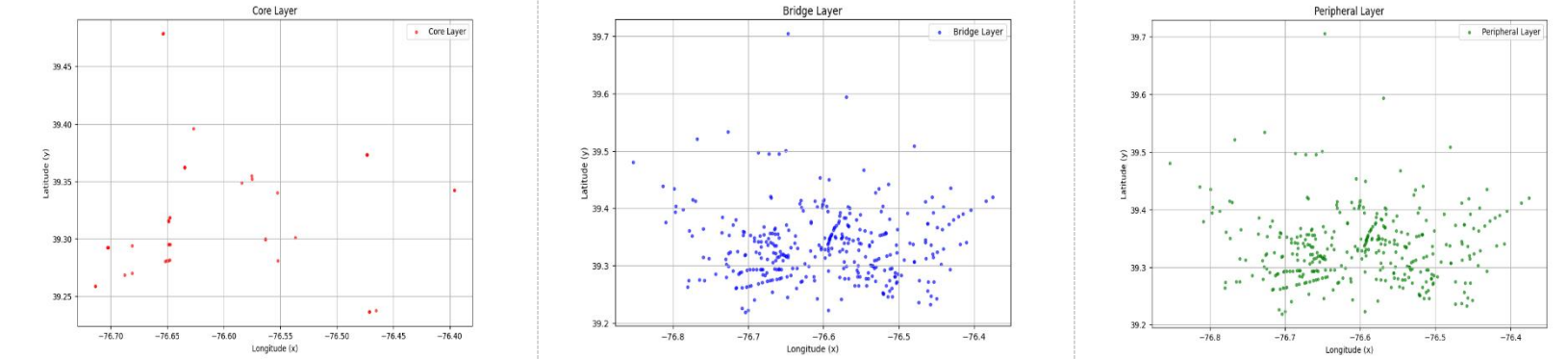


(b)Bridge Layer



(c)peripheral layer

Figure 12: Three-layer topology of a high-order transit network (for tourist)



(a)Core Layer (b)Bridge Layer (c)peripheral layer

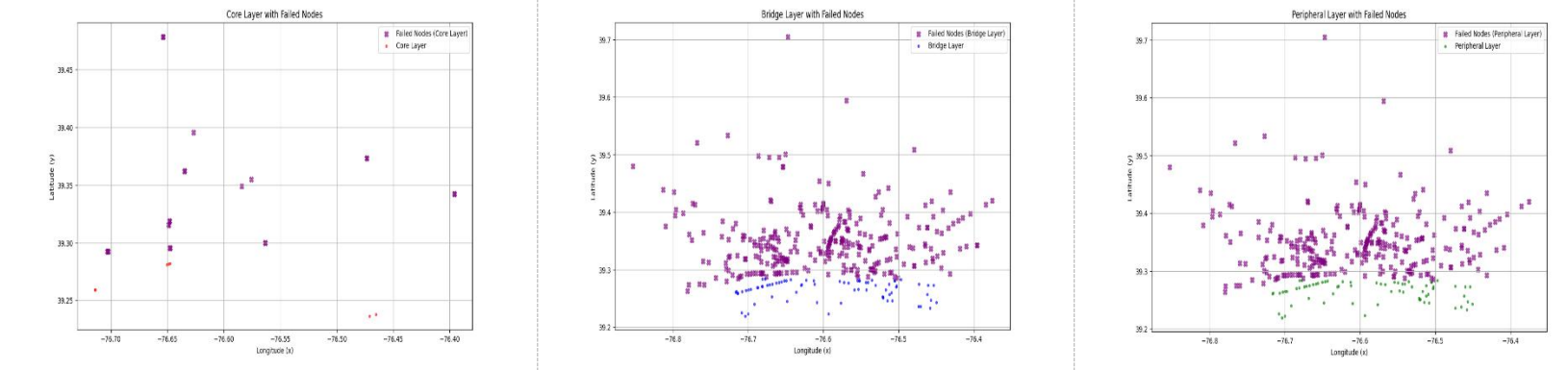
Figure 13: Three-layer topology of a high-order transit network (for suburban resident)

Based on the above hierarchical diagram, the preferences of different interest groups can be derived, and at the same time provide a good basis for our assessment of the bridge collapse.

6.3 the Collapse & Reconstruction

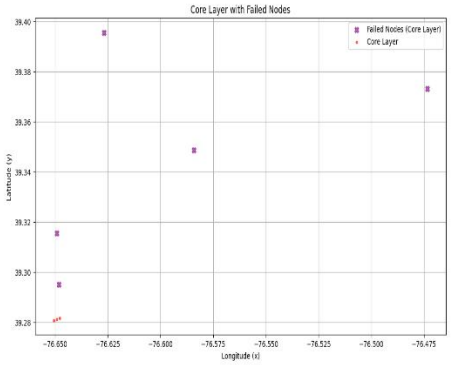
To simulate the impact caused by the collapse of the bridge, we need to select suitable datums and attack them in order to analyze the impact. Therefore, we can start by selecting two appropriate nodes on the map close to the bridge and use them as a datum to find the other nodes radiating from their center. (Requires a longitude between -76.72749° and -76.39875° and a latitude between 39.15056° and 39.28489°)

Subsequently, we attacked the selected set of nodes and fitted the remaining failed nodes that were affected, resulting in loads exceeding their carrying capacity through a cascading failure model, and visualized them into a three-tier transportation network corresponding to different interest groups to analyze the impacts of the bridge collapse on the local transportation ecosystems as well as on different segments of the population. The results are shown in the figures below (the crosses indicate the failed nodes).

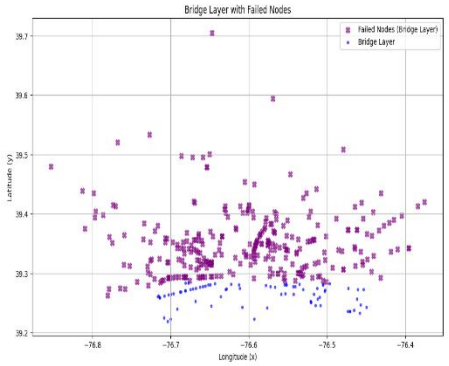


(a)Core Layer (b)Bridge Layer (c)peripheral layer

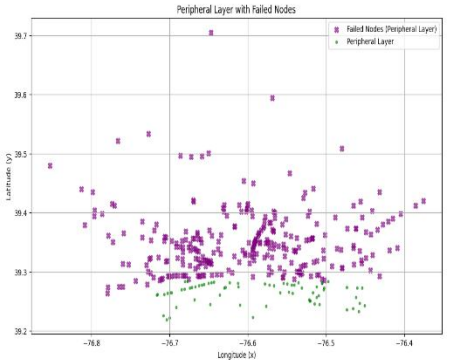
Figure 14: Three-layer topology after bridge collapse (for city resident)



(a)Core Layer

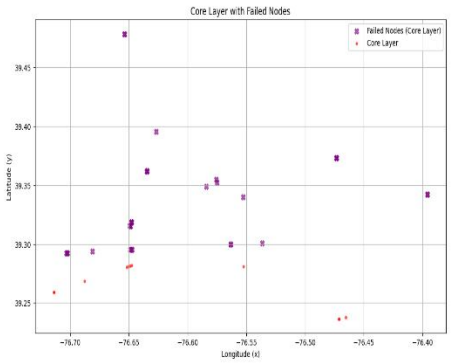


(b)Bridge Layer

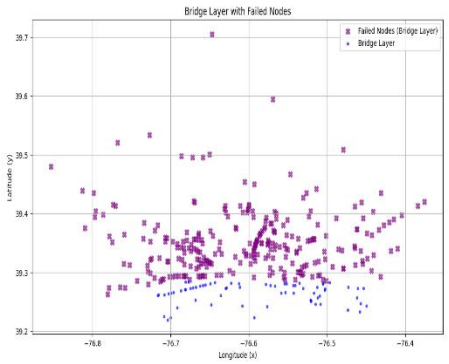


(c)peripheral layer

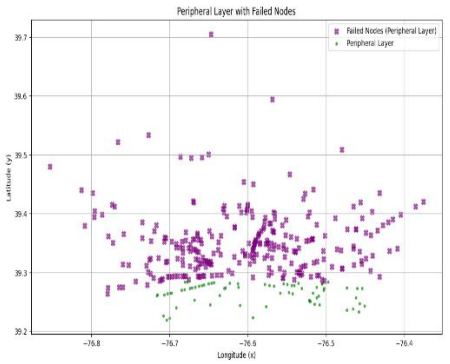
Figure 15: Three-layer topology after bridge collapse (for commuter)



(a)Core Layer

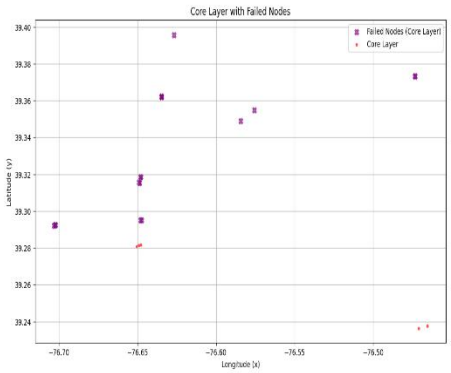


(b)Bridge Layer

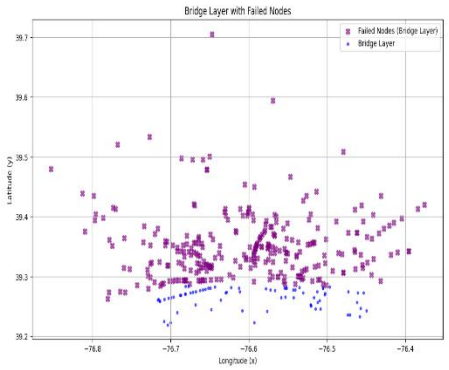


(c)peripheral layer

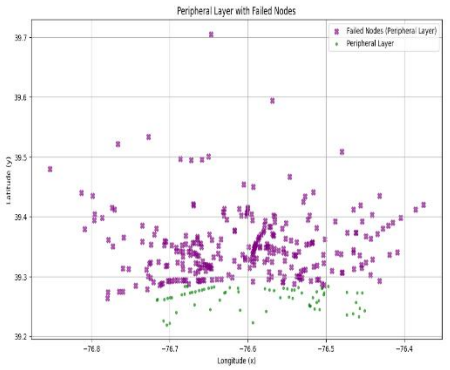
Figure 16: Three-layer topology after bridge collapse (for entrepreneur)



(a)Core Layer

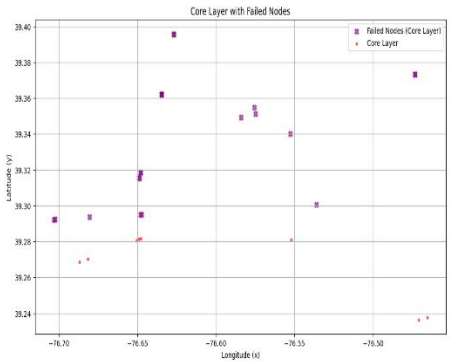


(b)Bridge Layer

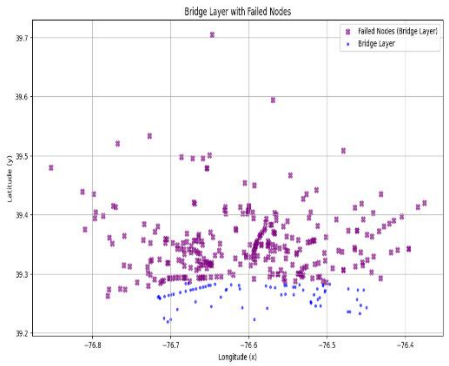


(c)peripheral layer

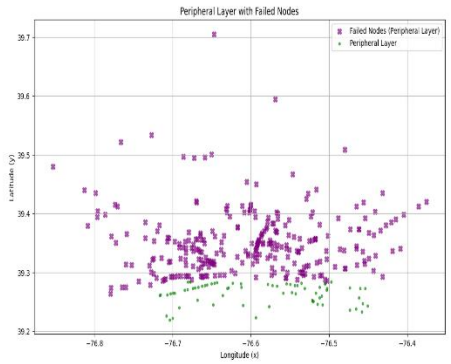
Figure 17: Three-layer topology after bridge collapse (for through traveler)



(a)Core Layer

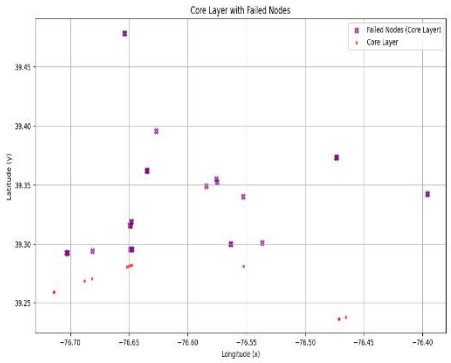


(b)Bridge Layer

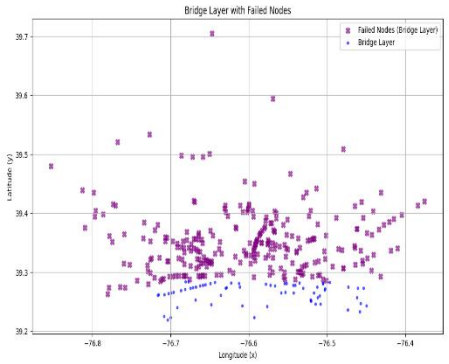


(c)peripheral layer

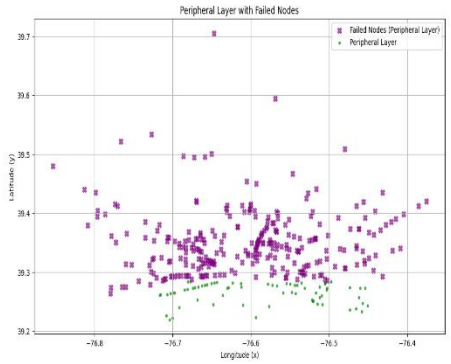
Figure 18:Three-layer topology after bridge collapse (for tourist)



(a)Core Layer



(b)Bridge Layer



(c)peripheral layer

Figure 19:Three-layer topology after bridge collapse (for suburban resident)

Based on the above image, the statistical data of different classes in the core layer are as follows:

Table 3: Core layer node failure rate

Classes	Total number of nodes	Total number of failed nodes	Nodal failure rate
City resident	21	16	76.2%
Commuter	8	5	62.5%
Entrepreneur	42	31	73.8%
Through traveler	22	17	77.3%
Tourist	29	21	72.4%
Suburban resident	42	30	71.4%

From the data in Table 3, it is easy to see that the collapse of the bridge had the greatest impact on urban residents and transit travelers, which can be interpreted in the following way: transit travelers rely on the bridge to continue their journeys across a particular region, and the collapse of the Francis Scott Key Bridge cut off their usual travel routes, resulting in the

disruption of a large number of related travel routes. The collapse of the Francis Scott Key Bridge cut off a large number of related travel routes. The impact of the bridge collapse on commuters is minimized, which can be interpreted as local commuters' daily commute is concentrated in a certain area within the city, and the collapse of the bridge has little impact on their daily travel paths. From the above analysis, it can be seen that the modeling has a certain degree of correctness and successfully fits the practical needs of various local stakeholders.

As for the bridge reconstruction, due to the high node failure rate at all levels, we can assume that the bridge plays an important role in the local transportation system and is somewhat irreplaceable. As a result, the local traffic situation can be restored to the normal level within a short period of time after the completion of the bridge reconstruction. In the process of reconstruction, the width of the bridge deck can be widened appropriately to enhance the throughput capacity of the local transportation system.

7 Task II: Project Analysis

In the context of optimizing Baltimore's public transportation network, the effectiveness of transit system connectivity is critical to determining the overall mobility and accessibility for urban and suburban residents. We chose to evaluate the impacts that a prolonged closure of US-40, a major highway, would have on the transportation network as well as community connectivity. Since public transportation is the backbone of daily commuting and economic activity, connectivity between different neighborhoods and transit stops plays a pivotal role in residents' daily lives and broader regional economic activity.

Given that downtown and suburban residents have very different travel patterns, it is necessary to assess how these differences affect their reliance on certain transit routes and transit stops. For downtown residents, downtown stops and routes connecting these areas are critical, while suburban residents are more reliant on inter-regional connections. We propose weighted network models and provide a framework for analysis that can assess the relative importance of nodes (bus stops) and edges (bus routes) based on these travel patterns, providing insight into the vulnerability of the network in the event of disruptions to critical routes.

Additionally, using a community detection algorithm based on modular optimization, we explore how the closure of Interstate US-40 could potentially disrupt the transit network, leading to isolated communities. These disruptions, especially in an interconnected transit system such as Baltimore's, could have a profound impact on transportation and affect the daily lives of residents, especially those in suburban areas who rely on long-distance routes.

7.1 Undirected Graph Modeling of Public Transportation Networks

In this section, since bus routes are reversible, we choose to model Baltimore's transit network as an undirected graph, where each bus stop serves as a node in the graph and each bus route serves as an undirected edge. Through this modeling approach, we are able to better understand the connectivity between bus stops and its impact on different stakeholders.

7.1.1 Definition of nodes and edges

Each bus stop can be considered as a node in the graph.The node characteristics n_i can be expressed as the following equation:

$$n_i = (\text{Site name, Coordinates, Type of service, } \cdots) \tag{1}$$

Each bus route can be considered as an undirected edge connecting the corresponding bus stops.The corresponding set of edges E can be expressed as follows:

$$E = \left\{ (n_i, n_j) \mid \text{Site } n_i \text{ and } n_j \text{ are connected by bus routes .} \right\} \tag{2}$$

7.1.2 Definition of weights

In order to more precisely characterize the importance of different sites and routes, we define weights for nodes and edges. The definition of weights will be adapted to the needs of different stakeholders.

Where the calculation of node weight w_{n_i} can be expressed by the following equation:

$$w_{n_i} = f(\text{Service Demand, Ridership, Transportation Connectivity,} \cdots) \tag{1}$$

Among other factors, service demand, ridership, and transit connectivity affect the weighting of each site. For example, downtown stops may have a higher weighting, while stops away from downtown have a lower weighting.

And the weights of the edges $w_{e_{ij}}$ can be expressed by the following equation:

$$w_{e_{ij}} = g(\text{Ridership, route length, frequency, } \cdots) \tag{2}$$

7.1.3 Inflows and outflows

The inflows and outflows at each stop can be calculated from actual traffic data, reflecting the amount of transit traffic entering or exiting the stop, using the following formula:

$$I_{n_i} = \sum_{j \in \text{Linked Neighbors}} w_{e_{ji}} \cdot \text{Traffic flow in border e}_{ji} \tag{1}$$

$$O_{n_i} = \sum_{j \in \text{Linked Neighbors}} w_{e_{ij}} \cdot \text{Traffic flow in border e}_{ij} \tag{2}$$

The travel patterns of different stakeholders may result in differences in flows that affect the amount of inflow and outflow at each site.

7.2 Community Detection Algorithm Based on Modular Optimization

We use Gephi's modular optimization algorithm can divide the nodes in the transit network into different communities based on the strength of connections between them. The specific steps

are as follows:

7.2.1 Community Detection

The association segmentation algorithm based on modularity optimization in GEPHI is used to cluster nodes into tight communities by calculating the connection strength between nodes. After substituting the undirected graph generated after assigning different weight values to different stakeholders, the following visualization results can be obtained as shown in Figure 20-21.

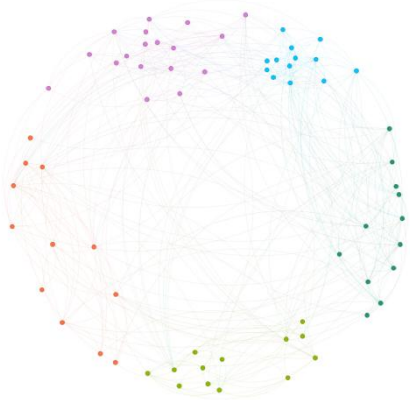


Figure 20: Community differentiation map for city residents

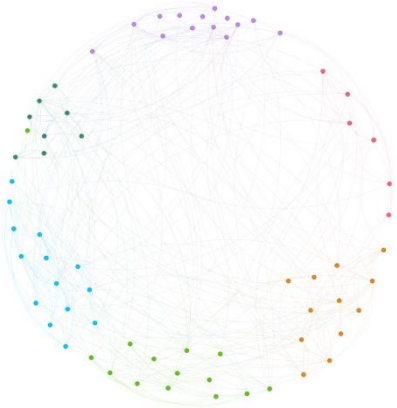


Figure 21: Community differentiation map for suburban residents

7.2.2 Impact analysis

Depending on the needs of different stakeholders, neighborhood segmentation can significantly affect the travel patterns of urban and suburban residents. For example, urban residents may be more likely to connect to downtown transit stops and routes that may be divided into one or more neighborhoods. Suburban residents, on the other hand, are more reliant on transit lines that connect to the suburbs, and these lines may form separate neighborhoods. Gephi's community differentiation map allows us to clearly see which bus routes are affected by neighborhood segmentation, thereby highlighting their key role in community connectivity. In particular, there are key routes whose closures or changes may significantly affect the flow of transit between communities and the travel choices of residents.

The recognition results are shown below:

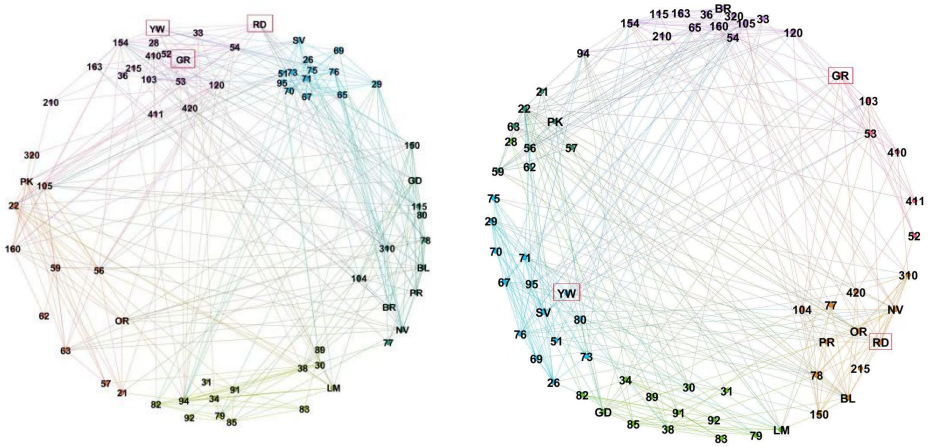


Figure 22: Results of identification of critical routes in city residential communities

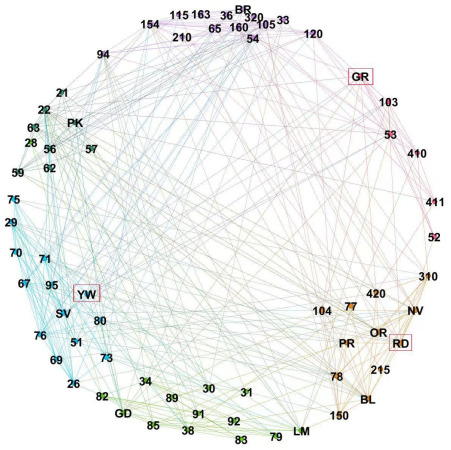


Figure 23: Results of identification of critical routes in suburban residential communities

7.2.3 Impact of the WBU program on community structure

From the community testing results, we can estimate the community-based impacts of the West Baltimore United (WBU) project on urban and suburban residents as follows:

Impacts on City Residents: For city residents, the WBU project resulted primarily in divisions within **Neighborhoods 1**. Residents within the community are significantly less connected, with fewer interactions between key transit stops in the city center. This fragmentation disrupts resident mobility in the urban core and may affect residents' access to central areas and key services.

Impacts on Suburban Residents: For suburban residents, the WBU project creates a split between **Neighborhoods 2, 3, and 5**, resulting in weaker connections between the different suburbs. Suburban residents face more transportation isolation and less interaction with each other, and suburban areas become more marginalized as a result. This reduced connectivity may limit suburban residents' travel options and affect their economic and social exchanges with other regions.

The estimation of these impacts allows us to identify areas where transportation connectivity is in dire need of enhancement. For example, enhancing connections between

different sub-communities within the city could be considered, along with improving transportation routes between suburbs to mitigate the negative impacts of the WBU project and restore overall connectivity to the affected areas.

8 Project Recommendation

8.1 Project Overview

1. Route Optimization: Based on the results of community testing, we propose to integrate 12 routes in the urban area with a duplication rate of more than 60%, and optimize them into 8 trunk routes. At the same time, five new suburban express routes will be added, adopting a “fast-slow” mode of operation, with 10-minute headways during peak periods and 20-minute headways during weekdays, focusing on connecting suburban residential areas with employment centers, sports stadiums, and other important destinations.

2. Reallocation of resources: The consolidation of routes in the urban areas will result in an annual savings of approximately $3 million in operating costs. These resources will be re-invested in the construction of suburban routes, including the deployment of new buses, the renovation of transfer facilities, upgrading information systems. Through scientific allocation of resources, the overall service level of the system will be improved.

3. System upgrading: optimize 30 key transfer hubs, add waiting facilities, electronic information screens, barrier-free access and other supporting facilities. Meanwhile, an intelligent scheduling system will be established to dynamically adjust the frequency of train departures according to real-time passenger flow data to improve the response efficiency of the system.

8.2 Benefits Analysis

8.2.1 Direct Benefits to Residents

With the optimization of urban transport, the travel situation of suburban residents will be significantly improved. The traffic flow model shows that their average commuting time will be reduced from 65 minutes to 42 minutes, and the return trip time during large events can be controlled within 45 minutes, with reduced travel time fluctuations on congested roads. The community detection model analysis shows that the average number of transfers for residents will be reduced from 2.3 to 1.1, with new direct routes covering a wide range of areas, and transfer hubs will be optimized and upgraded. In addition, five suburban express routes have been put into operation, with a 65% increase in peak capacity and a 45% expansion in service area, connecting many key areas and greatly enriching the travel choices of suburban residents .

8.2.2 Direct Benefits to Business

During the construction and renovation period, about 250 commercial tenants along the line will be affected but a compensation plan can be formulated. After the optimization of the line network, it can enhance the accessibility of the business district along the line in the medium term, increase the flow of passengers and turnover, form a commercial center, and add new commercial outlets and jobs, and in the long term, it can accelerate the integration of the

suburban economy, enhance the value of the land, attract the projects to the ground, form the pattern of linkage development, and promote optimization of the industrial layout in order to achieve the coordinated development of the region.

8.2.3 Direct Benefits to Transportation

Line optimization will increase the walking distance of about 15% of urban residents by 100 - 200 meters, concentrated in the area of 12 station mergers and 8 line adjustments, special groups are more affected, and improvement measures have been formulated for key areas; the new system will take 1 - 2 months for residents to adapt to the new system at the initial stage of operation, which involves changes in routes, timetables, and transfer modes, and publicity and guidance need to be strengthened; the transformation of 30 key stations will be carried out over a period of about 45 days in batches. The renovation of 30 key stations will take about 45 days and be carried out in batches. During the construction period, some stations will be temporarily adjusted, which will cause inconvenience for waiting and transferring, and some sections of the road will also be subject to traffic control, affecting the movement of social vehicles.

References

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[2] Huo Feizhou, Qu Fangxin, Ma Yaping, et al. Cascading failure model on directed weighted network of urban subway[J]. Journal of Safety Science and Technology, 2024,20(12):143-149.

[3] Xue Feng, He Chuanlei, Huang Qian. Identification of key nodes in Chengdu metro network and analysis of network performance[J]. China Safety Science Journal, 2019,29(01):93-99.

[4] Xie Fengjie, Wei Mengchan, Huang Jiaxin, et al. Identification and optimization of important bus lines based on a higher-order network modelS[J/OL]. Journal of Beijing Jiaotong University,1-12[2025-01-28].

The Necessary Path to Happiness

Dear Mayor,

We will write an overview of our model, emphasizing our use of historical data to refine the topology and presentation of the model. The model is intended to abstractly simulate the local Baltimore transportation network routes and to simulate the throughput capacity of the roadways based on data related to traffic flow.

In order to strengthen the reliability and accuracy of the model, a large amount of data was collected and the collected data were fully processed, including information related to different local nodes and roads in Baltimore. Considering that traffic route data are typical sequence data with time-dependent characteristics, we chose to construct a higher-order network model. In addition, we incorporate a cascading failure model to reliably quantify the importance of nodes, and generate a three-layer topology of the local high-order transportation network based on the quantification results to highlight the core demands of various stakeholders.

Based on our high-order network transportation model and our vision for the city's transportation planning, after model fitting and analysis, we provide two scenarios for local transportation planning that will further enhance the well-being and cohesion of the local community while ensuring the safety of the residents as well as the tourists traveling to and from the city.

- ✧ **Project 1: Reduce some of the duplicative short-distance routes in the city and add new inter-regional bus rapid transit routes.**

Benefits: The project will significantly improve the commuting efficiency of suburban residents, especially during high-demand hours, shorten their commuting time, relieve traffic pressure in the downtown area, and enhance the operational efficiency of the entire transit system.

Drawbacks: Some residents in the downtown area may be inconvenienced by the reduced number of short routes, especially those in low-density areas that rely on these routes. The initial implementation of the new system may require some adjustment time, resulting in short-term traffic instability.

- ✧ **Project 2: Upgrade downtown walkways and add street crossings.**

Benefits: The project will create a safer walking environment, encourage the public to use walking to get around, improve the overall health of residents, and provide more convenience and safety for people with disabilities.

Drawbacks: The construction of the project may cause short-term congestion and inconvenience to existing traffic flows. The high cost of infrastructure may lead to financial pressures on municipalities, especially in times of budgetary constraints.

Based on our modeling and consideration of the parameters, we are reasonably confident that the two projects recommended above are the path to happiness in Baltimore. This is because we have made every effort to balance the physical needs of the people with the safety of the projects themselves.

We look forward to the opportunity to work with you to delve deeper into our model and address any questions or concerns you may have.

Yours Sincerely,
Team # 2504020